

20.7 Summary and Notes

20.7.1 Summary

Idea.

1. Residuals are the projection of the errors onto $\mathcal{C}(\mathbf{X})^\perp$, with variances $\sigma^2(1 - h_{ii})$. The internally studentized residual r_i has a scaled beta law and is bounded by $\sqrt{n - p}$; the externally studentized residual t_i uses a scale estimated without case i (Definition 20.1.1 and Proposition 20.1.1).
2. Residual plots show only the part of a misspecified mean that is orthogonal to $\mathcal{C}(\mathbf{X})$: they can reveal wrong fitted values, not biased coefficients. The added-variable and component-plus-residual plots share their line and their residuals and differ in the horizontal axis (Proposition 20.1.2 and Proposition 20.1.3).
3. Leverage is potential influence. Under a Gaussian design $h_{ii} - 1/n$ is a scaled beta variable, so the $2p/n$ rule fires at a rate that depends on p . With finite second moments the largest leverage vanishes as n grows; with heavy-tailed regressors it need not (Proposition 20.2.1 and Proposition 20.2.2).
4. Partial leverage isolates the pull of a case on one coefficient, and h_{ii} is the leverage without $\mathbf{x}_j$ plus the partial leverage for $\mathbf{x}_j$ (Definition 20.2.1 and Proposition 20.2.3).
5. Deleting one case changes the coefficients, fitted values and residual variance by explicit formulas in $\hat{\varepsilon}_i$ and h_{ii} , and PRESS overestimates fixed-design prediction error slightly (Theorem 20.3.1 and Proposition 20.3.2).
6. t_i is the t statistic of the mean-shift outlier model and has the $t(n-p-1, \Delta\sqrt{1-h_{ii}}/\sigma)$ law; the maximum over cases, with a Bonferroni correction, is an outlier test of level at most α (Theorem 20.3.3).
7. Cook's distance, DFFITS and DFBETAS are products of a residual and a leverage factor. DFFITS is the largest standardized change in any linear combination of the coefficients. Their conventional cut-offs are scaled conventions, not tests (Definition 20.4.1, Theorem 20.4.1 and Proposition 20.4.2).
8. A set of cases can be deleted exactly, and the mean-shift F test checks a prespecified set (Theorem 20.5.1).
9. A cluster of m similar bad cases masks itself: each has leverage below $1/m$, its residual shrinks by $1 + mh_\star$, single deletion recovers a fraction $1/[m(1 + (m-1)h_\star)]$ of the group effect, and clean cases are swamped (Proposition 20.5.2).
10. Huber and bisquare M-estimates are computed by IRLS, which never increases the objective with a fixed scale. They resist outlying responses at cases of moderate leverage, but not bad leverage points. High-breakdown fits such as LTS, computed by concentration steps, expose clusters that least squares and M-estimates absorb (Proposition 20.6.1, Proposition 19.4.2 and Proposition 20.6.2).

20.7.2 Notes and sources

Sources. Seber and Lee (2003, sections 10.2 and 10.6) derive the studentized residuals and their laws, the leave-one-out diagnostics including COVRATIO and the Andrews–Pregibon statistic,

the mean-shift test for a set of cases, and brief accounts of masking, swamping and robust residuals. Christensen (2020, sections 12.1 and 12.5–12.6) develops leverage from the projection $\mathbf{M}$ and derives the updating formulae and predicted residuals. Sen and Srivastava (1990, chapter 8) and Rencher and Schaalje (2008, chapter 9) are briefer. The proofs here take the book's own route: the mean-shift model and the Frisch–Waugh–Lovell theorem for the law of t_i and for group deletion, orthogonal projections for the law of r_i , and partial leverage for DFBETAS.

Residuals and influence. Hoaglin and Welsch (1978) proposed the $2p/n$ rule. The t law of the externally studentized residual is due to Beckman and Trussell (1974). Cook (1977) introduced his distance and its ellipsoid reading, Belsley, Kuh and Welsch (1980) DFFITS, DFBETAS and COVRATIO with their size-adjusted cut-offs, and Andrews and Pregibon (1978) a determinant-based measure for sets of cases. Book-length treatments are Cook and Weisberg (1982), Atkinson (1985) and Chatterjee and Hadi (1988); the review of Beckman and Cook (1983) surveys outlier testing, including the Bonferroni approach. PRESS is from Allen (1974), and partial residual plots were popularized by Larsen and McCleary (1972). Proposition 20.2.1 is the regression form of the classical beta law for the Mahalanobis distance of a normal observation from the sample mean. Kiefer and Wolfowitz (1960) founded optimal design, which this book does not develop.

Masking and robust fitting. Beckman and Cook (1983) discuss masking and swamping at length; Proposition 20.5.2 is a formal version for the simplest configuration. Hadi and Simonoff (1993), Atkinson and Riani (2000) and Rousseeuw and van Zomeren (1990) give the procedures mentioned in [Section 20.5](#). M-estimation began with Huber (1964); Huber (1973) treated regression, under the leverage condition of Proposition 20.2.2. The bisquare is from Beaton and Tukey (1974), IRLS for robust regression was studied by Holland and Welsch (1977), and the finite-sample breakdown point is from Donoho and Huber (1983). LMS and LTS are from Rousseeuw (1984), with Rousseeuw and Leroy (1987) the standard account and Rousseeuw and Van Driessen (2006) the concentration step. The MM-estimator is Yohai's (1987). Maronna, Martin and Yohai (2006) give the theory; median and quantile regression are the subject of Chapter 45.

Data. Engel's budgets were made widely available by Koenker and Bassett (1982) and are distributed with statsmodels; the state data are those of [Chapter 6](#).

20.7.3 References

- Allen, David M. (1974). The Relationship Between Variable Selection and Data Augmentation and a Method for Prediction. *Technometrics* 16(1), 125–127.
- Andrews, David F. and Pregibon, Daryl (1978). Finding the Outliers that Matter. *Journal of the Royal Statistical Society, Series B* 40(1), 85–93.
- Atkinson, Anthony C. (1985). *Plots, Transformations and Regression: An Introduction to Graphical Methods of Diagnostic Regression Analysis*. Oxford: Clarendon Press.
- Atkinson, Anthony C. and Riani, Marco (2000). *Robust Diagnostic Regression Analysis*. New York: Springer.
- Beaton, Albert E. and Tukey, John W. (1974). The Fitting of Power Series, Meaning Polynomials, Illustrated on Band-Spectroscopic Data. *Technometrics* 16(2), 147–185.
- Beckman, Richard J. and Cook, R. Dennis (1983). Outlier.....s. *Technometrics* 25(2), 119–149.
- Beckman, Richard J. and Trussell, H. Joel (1974). The Distribution of an Arbitrary Studentized Residual and the Effects of Updating in Multiple Regression. *Journal of the American Statistical Association* 69(345), 199–201.
- Belsley, David A., Kuh, Edwin and Welsch, Roy E. (1980). *Regression Diagnostics: Identi-*

- tifying Influential Data and Sources of Collinearity*. New York: Wiley.
- Chatterjee, Samprit and Hadi, Ali S. (1988). *Sensitivity Analysis in Linear Regression*. New York: Wiley.
 - Christensen, Ronald (2020). *Plane Answers to Complex Questions: The Theory of Linear Models*. 5th edition. Cham: Springer.
 - Cook, R. Dennis (1977). Detection of Influential Observation in Linear Regression. *Technometrics* 19(1), 15–18.
 - Cook, R. Dennis and Weisberg, Sanford (1982). *Residuals and Influence in Regression*. New York: Chapman and Hall.
 - Donoho, David L. and Huber, Peter J. (1983). The Notion of Breakdown Point. In P. J. Bickel, K. A. Doksum and J. L. Hodges (eds.), *A Festschrift for Erich L. Lehmann*, 157–184. Belmont, CA: Wadsworth.
 - Hadi, Ali S. and Simonoff, Jeffrey S. (1993). Procedures for the Identification of Multiple Outliers in Linear Models. *Journal of the American Statistical Association* 88(424), 1264–1272.
 - Hoaglin, David C. and Welsch, Roy E. (1978). The Hat Matrix in Regression and ANOVA. *The American Statistician* 32(1), 17–22.
 - Holland, Paul W. and Welsch, Roy E. (1977). Robust Regression Using Iteratively Reweighted Least-Squares. *Communications in Statistics – Theory and Methods* 6(9), 813–827.
 - Huber, Peter J. (1964). Robust Estimation of a Location Parameter. *The Annals of Mathematical Statistics* 35(1), 73–101.
 - Huber, Peter J. (1973). Robust Regression: Asymptotics, Conjectures and Monte Carlo. *The Annals of Statistics* 1(5), 799–821.
 - Kiefer, Jack and Wolfowitz, Jacob (1960). The Equivalence of Two Extremum Problems. *Canadian Journal of Mathematics* 12, 363–366.
 - Koenker, Roger and Bassett, Gilbert (1982). Robust Tests for Heteroscedasticity Based on Regression Quantiles. *Econometrica* 50(1), 43–61.
 - Larsen, Wayne A. and McCleary, Susan J. (1972). The Use of Partial Residual Plots in Regression Analysis. *Technometrics* 14(3), 781–790.
 - Maronna, Ricardo A., Martin, R. Douglas and Yohai, V´ictor J. (2006). *Robust Statistics: Theory and Methods*. Chichester: Wiley.
 - Rencher, Alvin C. and Schaalje, G. Bruce (2008). *Linear Models in Statistics*. 2nd edition. Hoboken, NJ: Wiley.
 - Rousseeuw, Peter J. (1984). Least Median of Squares Regression. *Journal of the American Statistical Association* 79(388), 871–880.
 - Rousseeuw, Peter J. and Leroy, Annick M. (1987). *Robust Regression and Outlier Detection*. New York: Wiley.
 - Rousseeuw, Peter J. and Van Driessen, Katrien (2006). Computing LTS Regression for Large Data Sets. *Data Mining and Knowledge Discovery* 12(1), 29–45.
 - Rousseeuw, Peter J. and van Zomeren, Bert C. (1990). Unmasking Multivariate Outliers and Leverage Points. *Journal of the American Statistical Association* 85(411), 633–639.
 - Seber, George A. F. and Lee, Alan J. (2003). *Linear Regression Analysis*. 2nd edition. Hoboken, NJ: Wiley.
 - Sen, Ashish and Srivastava, Muni (1990). *Regression Analysis: Theory, Methods, and Applications*. New York: Springer.
 - Yohai, V´ictor J. (1987). High Breakdown-Point and High Efficiency Robust Estimates for Regression. *The Annals of Statistics* 15(2), 642–656.